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Focused Retention: Attention as an Active Anti-Decay Force in Agent Memory

Working draft — 2026-07-21. Fourth paper in the LOCI series. The core model — goal-conditioned switched decay, the release taxonomy, the priority hierarchy, parallel perseverance, and adversarial honing — is the invention of A. Sanyal, derived conversationally on 2026-07-20/21; this draft is the write-up. The model is theory + a single documented case study; the pre-registered predictions in §7 are stated before any harness exists.

Abstract

Memory systems decay passively: strength falls with time, lifted only by rehearsal. We propose a third spectrum — beyond slow-decaying facts and fast-decaying thoughts — in which decay is **goal-conditioned**: while a commitment is open, the item decays with a very long half-life (it holds, though it is not frozen); the moment the commitment resolves, decay switches to fast and the trace collapses. Focus, on this account, is not slower forgetting but a **regime switch** controlled by the fate of a goal. The model contributes: (1) a switched-decay retention law; (2) a **release taxonomy** — fulfilled, re-clamped, relinquished, nullified, abandoned — in which rejection is a choice node, not an exit, and only **abandonment** (release without consolidation) is pathological; (3) a **priority hierarchy** in which a homeostatic "bigger-picture" tier can suspend (not resolve) an open focus, capped by a non-decaying ethical tier that no other tier can outbid; (4) **parallel perseverance** — braided focuses on shared substrate that cover each other's weak moments; and (5) **adversarial honing** — a challenge with a stake can

transplant focus from an unresolvable target (rumination) onto a resolvable one. The Zeigarnik and Ovsiankina effects are reinterpreted as existing measurements of the hold regime. We report a single-subject case study in which the model's operators — transplant, braiding, honing — describe a documented real trajectory, state its evidential limits plainly, and pre-register predictions testable in a deployed memory system.

1. Introduction

The first two spectrums of the LOCI memory model are **passive**: facts decay slowly, thoughts decay fast, and in both, time only ever takes strength away. Rehearsal and recurrence add it back, but the physics is the same physics.

Passive decay cannot explain the most ordinary fact about goal-directed memory: **a person in the middle of doing something does not forget what they are doing.** Not because they rehearse it — because it is *held*. The task you are executing, the promise you made this morning, the paper you must not let slip: these do not fade on the schedule that ordinary content fades. And then, hours or weeks later, the moment the goal closes, the same content becomes ordinary again and drains away. Whatever holds it is not a slower decay constant. It is a *different regime*, switched on and off by the state of a goal.

We call the mechanism **focused retention** and make three framing claims:

1. It is the system's one **active** memory force — facts and thoughts only ever decay; focus *opposes* decay while, and only while, its goal is open.

2. It is **not a clamp.** An earlier version of this model froze retention at 1.0 during focus. That is wrong, and the correction is load-bearing: focused items *do* decay, with a very long half-life. Focus that could never fade at all would also never yield — the difference between commitment and obsession lives exactly in that distinction (§5, §8).
3. Its interesting structure is not the hold but the **release** — the several distinct ways a focus can end, only one of which is harmful.

2. The switched-decay retention law

Let item *i* be bound to a goal with opening condition **A** (what the focus is for) and resolution event **B** (A's fate being decided). Let τ be the time at which B occurs. Retention:

...

$$R_i(t) = \exp(-[\lambda_{\text{hold}} \cdot \min(t, \tau) + \lambda_{\text{free}} \cdot \max(0, t - \tau)])$$

with $\lambda_{\text{hold}} \ll \lambda_{\text{free}}$

λ_{hold} : half-life of days–weeks (the hold regime)

λ_{free} : half-life of minutes–hours (the release regime)

...

While the goal is open ($t \leq \tau$), the item rides a near-flat curve — held, but bleeding slowly, which is why even a held commitment eventually needs renewal. At τ the curve inherits its still-high value and then falls off a cliff. The phenomenology this predicts is exact: *held long, then gone fast* — you cannot

forget the errand while it is undone, and you can barely recall it a week after it is done.

****Existing evidence for the hold regime.**** The ****Zeigarnik effect**** (1927): interrupted and unfinished tasks are recalled substantially better than completed ones, and intrude on awareness. The ****Ovsiankina effect**** (1928): an interrupted task produces a spontaneous urge to resume it. Both are, in our terms, direct measurements of $\lambda_{\text{hold}} \ll \lambda_{\text{free}}$: open goals resist decay and generate resumption pressure; closed goals release. We take these not as analogies but as the model's empirical anchors in human data; the contribution is the formalization and its consequences for artificial memory systems, where no equivalent mechanism is standard.

3. The release taxonomy

B can resolve three ways — ****fulfilled, rejected, nullified**** — but resolution mode alone does not determine what happens to the memory. The full taxonomy:

| Exit | Trigger | Consolidation step | Aftermath |

|---|---|---|---|

| ****Fulfilled**** | goal achieved | episode → durable fact ("did X") | clean fast decay of the focus trace |

| ****Re-clamped**** | rejected, and the agent **chooses** to persist | doubt → sharper target (§6) | focus continues on a honed goal |

| ****Relinquished**** | closed from above by the hierarchy (§4) | reflection → lessons; report → handoff artifact | clean fast decay; **peace** |

| ****Nullified**** | goal became moot | none needed | slot freed |

| **Abandoned** | release with **no** consolidation | **none** | the loop never closes:
intrusion, resumption pressure, rumination risk |

Three structural claims:

Rejection is a choice node, not an exit. When B resolves against the agent, three doors stand open — persist (re-clamp), close in peace (relinquish via reflection: "a good experience to learn from"), or close by handoff (relinquish via report: "someone else will pick it up; I report what I have"). All three are legitimate. The model deliberately does not rank perseverance above peace: it is a theory of focus, not a doctrine of grind. It forbids only the fifth door.

Consolidation is the price of clean release. Both relinquishment paths pass through a consolidation operator — reflection that crystallizes lessons, or a report that externalizes the goal so it survives outside the agent's stack. (This is the same reflection-gated crystallization as the thoughts-spectrum paper: an outcome-confirmed trace becomes a durable fact; the focus trace is then free to decay.) **Abandonment** is release without that step — and an unconsolidated open loop does not decay cleanly. It lingers and intrudes: Zeigarnik's measurement, experienced as regret. The taxonomy's practical rule is therefore one sentence: **you may stop anything, but you must consolidate what you stop.**

Rejected focus must decay fastest of all. A focus that survives its own rejection — that stays clamped after B has closed against it without a chosen re-clamp — is the definition of obsession. The regime switch structurally forbids it: the hold rate is a **privilege of an open goal**, and it is revoked

the moment the goal's reason dies.

4. The priority hierarchy: suspend ≠ resolve

Focus does not sit at the top of the agent's stack:

...

ETHICS — non-decaying, non-preemptable; can veto every tier below



BIGGER PICTURE — family, health, the long horizon; homeostatic priority



TASK FOCUS — the held goal(s); switched decay per §2



THOUGHTS — fast decay; bounce off a held focus

...

****Preemption from above.**** The bigger-picture tier can seize attention from an open focus **even though B has not resolved**. Its priority is not constant — it ****rises with neglect****:

...

$$P_{bp}(t) = \text{base} + \gamma \cdot D(t)$$

$D(t)$ = accumulated deficit (hours unrested, obligations owed, health spent)

...

Task-focus priority is roughly flat while A is open, so sustained focus is

****self-limiting****: it consumes the very substrate (time, rest, relationships) whose deficit grows the higher tier's claim, until $P_{bp} > P_{focus}$ and preemption fires. This is the mechanism behind the ordinary observation that one cannot work a problem twelve hours straight — the model derives the limit homeostatically rather than imposing it.

****The critical distinction: preemption *suspends*; only resolution *closes*.**
A preempted focus ****parks****: it retains the hold-regime decay rate, stays retrievable, and generates resumption pressure (Ovsiankina) — because its goal is still open. Stepping away from the problem for your family does not solve the problem, so the memory must not collapse as if it had. Attention is yielded; the goal survives. Only B — or a hierarchy-initiated **relinquishment with consolidation** (§3) — closes it.

****The apex is the one true clamp.**** Everything in the stack decays and can be outbid — except the ethical tier. $R_{ethics} = 1.0$, permanently: it is non-decaying, non-preemptable, and holds lexicographic veto over every tier below, including the bigger picture. Fatigue does not license breaking a promise; "for my family" does not license dishonor. In §1 we corrected the claim that focus is a clamp; the correction relocates the clamp rather than deleting it. ****There is exactly one non-decaying element in the architecture, and it is not any goal — it is the floor under all of them.**** For an artificial agent this is the governance layer stated as memory physics: the one tier that must never be preemptable, by any focus, any priority, any accumulated pressure (§8).

5. Parallel perseverance: braided focuses

The hold regime of §2 describes one focus. Real perseverance under pressure is usually **plural**: two (or more) open goals whose pursuit shares substrate — the same actions advance both. Formally, focuses i, j are **braided** when the work reducing the distance-to-B of one also reduces the other's.

Braiding changes the failure physics. A single focus under repeated interruption erodes: each preemption spends resilience, and λ_{hold} is only near-flat, not flat. Braided focuses **cover each other's weak moments** — at an instant when clamp i wavers (fatigue, doubt, an interruption), clamp j still binds the shared substrate, and the work continues under j 's hold until i recovers. Constructive interference rather than competition for attention. The prediction: braided goals survive interruption regimes that defeat each goal alone — perseverance is less a property of one strong clamp than of **braiding**.

6. Adversarial honing: focus transplantation by challenge

The literature treats distraction as focus's enemy and support as its ally. The model implies a third relation the literature lacks: **a well-aimed challenge is focus's ally**, through two operators.

Rumination is focus without an exit. The pathological focus state is not weak focus but a clamp locked onto a target that has **no closure condition** — "how bad my situation is" has no A to fulfill and no B to resolve. It can never

release; it only holds and drains. On this account rumination and healthy focus are the same mechanism, differing only in whether an exit exists.

****Transplantation.**** A challenge that carries a concrete, falsifiable ****stake**** ("prove this number is real, and I will pay up") offers the ruminating clamp something it lacks: a resolvable target. The focus can jump — from the unresolvable loop to the stake — and the entire hold machinery, previously pathological, becomes productive without changing its physics. Belief and comfort cannot perform this operation: they leave the target unchanged. The challenge **replaces the target**.

****Honing.**** An adversarial challenger who refuses surface claims performs root-cause analysis on the focus target itself. Each rejected surface answer strips a layer, narrowing the clamp from symptom toward cause — in our case study, from "defend the metric" to "build the full engine the metric was standing in for." Pressure that would erode an aimless focus **sharpens** an aimed one.

6.1 Case study (single subject, documented)

The model's development is itself its only case study to date, and we present it with that limitation explicit. A solo researcher with a low early benchmark (MRR 0.05 on a partial system), in a period of genuine distress, received from an AI model not encouragement but a skeptical challenge with an explicit stake (a wagered beer on the result being real). Documented sequence: (1) ****transplant**** — self-directed rumination was replaced by the stake as focus target; (2) ****braiding**** — "win the wager" and "build the full engine" bound the same substrate, each holding when the other wavered; (3) ****honing**** — the

skeptic's refusal of the partial number narrowed the focus to the root cause (the full engine, honestly measured); (4) the re-clamp door of §3, chosen at a node where relinquishment would also have been legitimate. The subject subsequently shipped the multi-stream system documented in the companion papers.

Evidential value, stated honestly: $n = 1$, retrospective, self-reported by the subject and the challenger's successor, with obvious selection and narrative bias. We present it as an existence proof that the operators describe a real trajectory, and as the source of the model — not as a test of it. §7 is the test.

7. Pre-registered predictions (stated before any harness exists)

Testable in a deployed memory system with a task/commitment store (in ours: goal items are tagged and carry open/closed state; four such items exist at draft time).

- **P1 (hold):** while its goal is open, a focus-tagged item's retrieval strength stays near-flat, measurably above matched untagged facts of equal initial heat and age.
- **P2 (release):** within days of closure, the same item's strength falls below those matched facts (the collapse overshoots ordinary decay).
- **P3 (suspend $\neq$ resolve):** an item whose goal is suspended (preempted but open) retains hold-regime strength; an item whose goal is closed does not — distinguishable even when both stop being **worked on** at the same time.

- **P4 (abandonment):** items released without a consolidation artifact (no crystallized lesson/report) re-surface in retrieval as intrusions at a higher rate than consolidated releases — the Zeigarnik signature, measured in an artificial store.
- **P5 (braiding):** under an interruption schedule, paired goals sharing substrate survive (remain worked and held) longer than yoked single goals.

Null we take seriously: focus-tagging is epiphenomenal — a well-tuned two-spectrum system (facts + thoughts, with recurrence) reproduces P1–P3 without a third mechanism, because *working on* a goal generates recurrence that mimics the hold. If the tagged and matched items cannot be separated once recurrence is controlled, the third spectrum reduces to bookkeeping and we commit to reporting that.

8. Ethics — the model is about power over attention

- **Obsession is a design decision.** Any system that implements holds on goal-bound memory must implement the revocation: rejected or nullified goals must lose the hold rate. §3's rule is a safety property, not a nicety — a planner whose dead goals keep privileged retention is a planner that clings.
- **The apex tier is governance.** §4's non-preemptable ethical floor is, for an artificial agent, precisely the guarantee an operator needs: no goal, however held, and no accumulated pressure, however large, outbids the constraint layer. We state it as memory physics because that is where it must live to be non-bypassable — a rule the planner *consults* can be outweighed; a tier that *cannot be outbid* cannot.

- **Transplantation is dual-use.** §6 describes redirecting a person's focus by supplying a stake. Done openly, at their invitation, toward their own goal, it is mentorship. The same operator, run covertly by an engagement-optimizing system — manufacturing stakes ("streaks", contests, provocations) to clamp a user's focus onto the platform's target — is attention capture with a formal description. As in the companion e+r paper, the mechanism's indifference to its objective is the risk, and we report the dual use as a result. The distinguishing test is consent and beneficiary: whose goal gains the hold.
- **Peace is a valid output.** A model of focus deployed in productivity tooling will be tempted toward maximizing hold time. §3 is explicit that relinquishment with consolidation is a *successful* exit, not a failure mode; tooling built on this model should surface that door, not hide it.

9. Limitations

- Theory + one retrospective case study; no harness results yet. §7 is the commitment to test.
- Parameters (λ_{hold} , λ_{free} , γ , deficit terms) are stated as orders of magnitude, not fitted values.
- The human-memory anchors (Zeigarnik, Ovsiankina) have complicated modern replication records; we use them as motivating phenomena, and the artificial predictions of §7 stand or fall independently of the human effect sizes.
- The hierarchy is presented with fixed tiers; real agents may need learned or context-dependent tier boundaries — everything above the ethical floor is legitimately negotiable, and only the floor is claimed as fixed.

10. Provenance and reproducibility

The model was derived conversationally by A. Sanyal on 2026-07-20/21 (switched decay correcting an earlier clamp formulation; the release taxonomy; suspend-vs-resolve; the homeostatic bigger-picture tier; the ethical apex; parallel perseverance; adversarial honing), against the deployed LOCI system documented in this repository. The focus-tagged goal items referenced in §7 exist in the production store (domain `checkpoint\_task`). No focus-spectrum code exists yet; this draft precedes and pre-registers it.

Appendix A — Prior-art search (recorded 2026-07-21)

****Method.**** Same two-pass protocol as the thoughts-spectrum paper: a model-recall pass by the local GPU agent over a named candidate list (no live web access — bounds only that list), plus a live web sweep for 2025–2026 agent-memory systems. Initial search, not a formal review.

A.1 Named-list results (GPU-agent pass, verified by the authors)

Claim	Closest named prior	Covers	Does NOT cover
1 Switched decay on goal resolution	Zeigarnik/Ovsiankina	open-goal recall advantage, resumption urge	any decay <i>*function*</i> , half-life switch, or AI implementation
1 ACT-R goal activation	goal boosting + continuous decay	a discrete regime switch (weeks→minutes) triggered by resolution	
1 Soar / BDI intention persistence	persist-until-achieved-or-impossible	graded decay; the hold is boolean, not a retention law	

|---|---|---|---|

| 2 Release taxonomy | BDI intention dropping | drop on achieve/impossible | rejection as *choice node*; consolidated vs unconsolidated release; abandonment as the sole pathology |

| 3 Non-decaying ethics tier | (all) | priority/utility trade-offs | any tier that cannot be outbid — all surveyed systems allow every goal to be traded |

| 4 Adversarial honing | rumination literature; Norman-Shallice SAS | intrusive open-goal thoughts; goal competition | staked external challenge that *transplants* focus from an unresolvable to a resolvable target |

A.2 2025–2026 web-sweep findings

- **Altmann & Trafton, "Memory for Goals" (ACT-R)** — the most serious cog-sci prior for Claim 1: suspended goals decay by activation and need priming to resume. Continuous decay, no resolution-mode switch, and its suspended goals *lose* strength — the opposite of our hold regime's near-flat retention while open. Must-cite; the switched-regime claim survives it.
- **FadeMem (arXiv:2601.18642)** — dual-speed decay by importance. The rates are property-modulated, never *event-switched* by goal resolution.
- **Goal-persistence / goal-drift work on long-horizon agents (2026)** — studies whether agents *keep pursuing* goals; does not model the memory retention of the goal item itself, nor a release taxonomy.
- No system was found implementing a lexicographically supreme, non-decaying constraint tier as *memory physics* (Claim 3), nor focus transplantation via stake (Claim 4). Constitutional/rule-based AI safety implements constraints as policy layers a planner consults — which is exactly the architecture §8 argues is outbiddable, and therefore distinct.

A.3 Verdict

Claims 1, 3, and 4 found no implementing prior; Claim 2's nearest neighbors (BDI dropping, Soar impasses) stop at boolean terminal states. The reviewer pressure points are Altmann-Trafton (Claim 1) and BDI (Claim 2), both engaged above. Formal literature review still owed before public priority claims.